



Ph.D. Scholar, Department of CSE
Indian Institute of Technology Patna
☎ (+91) 8532045278
✉ er.naseembabu@gmail.com
in LinkedIn

Naseem Babu

Education

- 2020–Present **Ph.D. in Computer Science & Engineering**, *Indian Institute of Technology, Patna.*
Research Area: Classification of Time Series Data
CGPA: 7.75
- 2018–2020 **M.Tech in Computer Engineering**, *Malaviya National Institute of Technology, Jaipur.*
Project: Mitigation of Sybil Attack in Mobile RPL Routing Protocol (Internet of Things)
CGPA: 7.38
- 2013–2017 **B.Tech in Information Technology**, *Dr. Ambedkar Institute of Technology for Handicapped, Kanpur.*
Project: Smart Home Automation using Google Assistant and Blynk App
Percentage: 73.32

Area of Interest

Machine Learning, Deep Learning, Artificial Intelligence, Brain-Computer Interface (BCI), Electroencephalography (EEG), Emotion Recognition, Biomedical Signal Processing, and Audio Signal Processing.

Publications

Journal Articles

- 2025 **Naseem Babu**, Udit Satija, Jimson Mathew, and A.P. Vinod. Emotion recognition in virtual and non-virtual environments using eeg signals: Dataset and evaluation. *Biomedical Signal Processing and Control*, volume 106, page 107674, 2025.
- 2025 **Naseem Babu**, Jimson Mathew, Udit Satija, and A. P. Vinod. Unsupervised domain adaptation for eeg-based emotion recognition in virtual environment. *SN Computer Science*, 2025. Under review.
- 2024 **Naseem Babu**, Dayananda Pruthviraja, and Jimson Mathew. Enhancing lung acoustic signals classification with eigenvectors-based and traditional augmentation methods. *IEEE Access*, volume 12, pages 87691–87700, 2024.

Conference Papers

- 2022 **Naseem Babu**, Jyoti Kumari, Jimson Mathew, Udit Satija, and Arijit Mondal. Multiclass categorisation of respiratory sound signals using neural network. In *2022 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, pages 228–232, 2022.
- 2022 **Naseem Babu**, Prattipati Kumar, Jimson Mathew, and Udit Satija. Exploration of bonafide and spoofed audio classification using machine learning models. In *2022 IEEE 19th India Council International Conference (INDICON)*, pages 1–6, 2022.

Skills and Strengths

- Languages Fluent in English and Hindi (Reading, Writing, and Speaking)
- Coding Python, Java, C
- Frameworks PyTorch, TensorFlow, Keras, Scikit-learn
- Libraries Pandas, NumPy, SciPy, Matplotlib, Seaborn, Librosa, MNE
- Platforms Google Colab, Jupyter Notebooks
- Control Tools Git, GitHub, LaTeX
- Strengths Hard-working, Quick Learner, Collaborative, Persuasive, Motivator, Leader

Academic & Work Experience

Teaching Assistant for CS1201: Data Structure, IIT Patna
Teaching Assistant for CS229: Computer Architecture, IIT Patna
Instructor, High-End Workshop on Deep Learning-Based Biomedical Systems, IIT Patna
Guest Lecturer, Parul University, Gujarat, India
Teaching Assistant, MNIT Jaipur

Academic Achievements & Honours

Grand Challenge Student Research Grant (BioCAS 2022) – Grand Challenge on Respiratory Sound Signal Classification, 2022
University Grants Commission (UGC) Fellowship for Ph.D. Research (UGC-NET JRF 2022 Qualified)
MHRD Fellowship for M.Tech Research Work
All India Rank (AIR) 1893 in GATE 2018 (3-Time GATE Qualified)

Personal & Professional Engagement

Ph.D. Placement Coordinator, IIT Patna
Avid Reader of Motivational and Self-Development Books
Fitness Enthusiast Walking and Health-Conscious Lifestyle
Passionate about Interacting with People and Building Connections
Passionate About Exploring Cutting-Edge Technologies

Referees

Prof. Jimson Mathew

(Ph.D. Supervisor)
Professor, Department of
Computer Science & Engineering
Indian Institute of Technology Patna, India
✉ jimson@iitp.ac.in

Prof. Vinod A Prasad

(Ph.D. Co-Supervisor)
Professor, Department of
Infocomm Technology
Singapore Institute of Technology, Singapore
✉ vinod.prasad@singaporetech.edu.sg